AE 481 Team 01 Assignment 9

November 17, 2025

1 Configuration Trades

Three key trades were carried out regarding the design of the F/A-67: mission performance and sizing, planform design, and external ordnance and fuel carriage.

1.1 Mission Performance and Sizing Trade

Aircraft wingspan, planform area, and aspect ratio were reanalyzed using refined weight estimation methods for empty weight prediction and fuel fraction. Additionally, detailed drag polars based on the geometry of the preliminary F/A-67 design were incorporated into the weight estimation and sizing plot routines. Because of the refined methods, the feasibility of designs for various mission performance parameters such as combat radius changed. The trade study was performed by varying combat radius, combat time, and dash Mach number for the air-to-air mission. The air-to-air mission was by far the more constraining mission type because of the supersonic dash and sustained-turn rate constraints. Essentially, a mission that was feasible for the air-to-air mission was feasible for the strike mission, but not the other way around. The objective analyzed for the trade study was aircraft unit cost. The specific requirement was for the unit cost to not exceed that of the F/A-18E/F (113 $M 2025). This enables higher performing aircraft despite the increasing cost.

In addition to varying the air-to-air mission parameters, the aircraft wingspan and aspect ratio were varied over ranges based on historical aircraft. Wingspan was varied from 45 ft to 60 ft, the maximum value, and aspect ratio was between 2.7 and 4. For selecting a design point, an additional goal was to reduce wingspan and planform area over the preliminary design of 60 ft and 670 ft^2. The result of the trade study was visualized using a matrix of lattice plots, with combat radius and combat time varied in individual carpet plots, Mach number varied over a lattice plot, and wingspan and aspect ratio varied over the matrix of plots. For clarity, only relevant lattice and carpet plots are included.

The wingspan and aspect ratio were determined by identifying the lattice plot with the largest feasible region while reducing wingspan and aspect ratio from the preliminary design. The purpose of this was to reduce planform configuration issues, primarily the 50 ft length limit of the aircraft. The final wingspan and aspect ratio selected were 48.75 ft and 3.67 respectively. Using the lattice plot, a dash Mach number of 1.65 was achieved. Figure 1 shows the carpet plot for the selected wingspan, aspect ratio, and dash Mach number. The blue hatched curve separates infeasible and feasible designs using the sizing plot and constraint curves. Feasible designs are on the opposite side of the hatch marks. For all designs analyzed, the unit cost does not exceed the 113 $M 2025 requirement. An interesting trend is that for moderate combat radii and combat times, the cost is nearly constant. Eventually, increasing combat time and combat radius yields a steady increase in cost.

The selected design point was determined based on achieving at least one desired mission performance

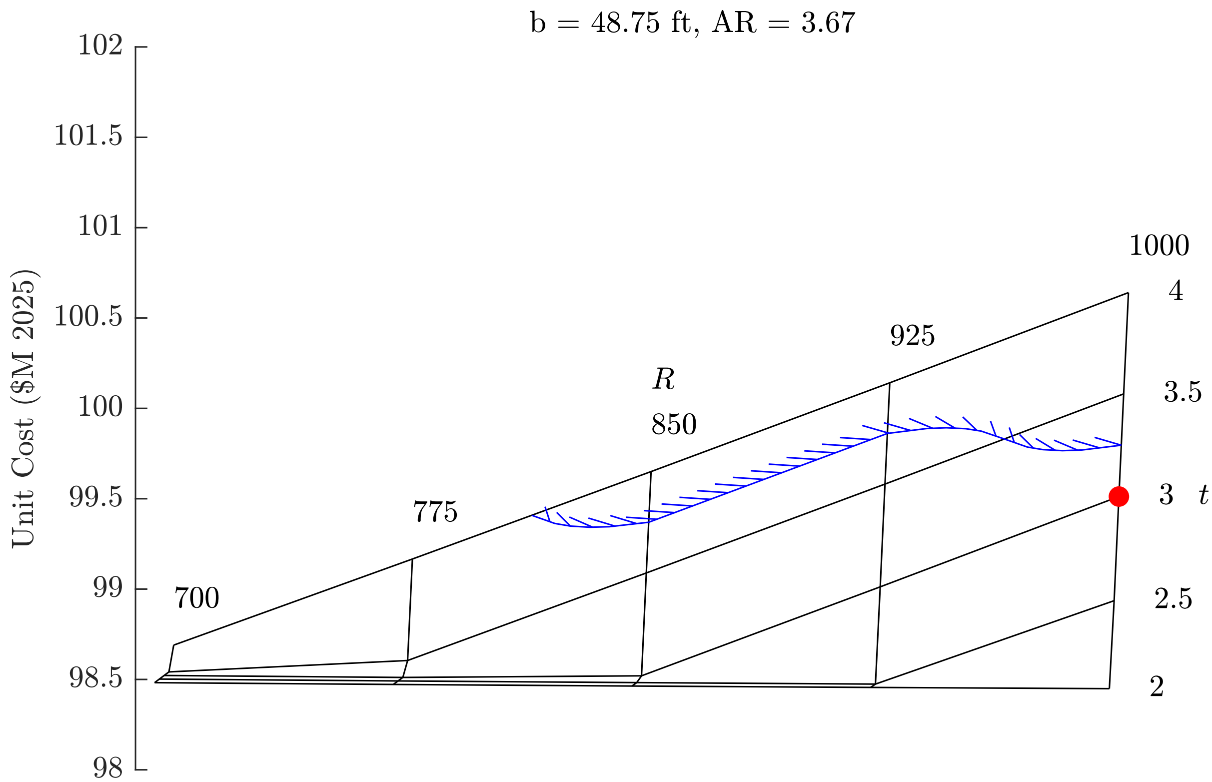

Figure 1: The carpet plot for dash Mach number of 1.65 varies combat radius in nmi and combat time in min. The blue hatched curve roughly indicates the boundary between feasible and infeasible designs. The selected design point is marked in red.

parameter. Additionally, all design points were analyzed for the desired sustained turn rate of 10 deg/s. In this case, the desired combat radius of 1,000 nmi can reasonably be reached without sacrificing combat time significantly. The desired combat time of 5 min is beyond the analyzed region of the carpet plot, but it can be extrapolated that nearly all of the combat radius would need to be sacrificed. Therefore, the selected mission performance is a 1,000 nmi combat radius, a 3 min combat time, a dash Mach number of 1.65, and a sustained turn rate of 10 deg/s. All parameters exceed the minimum requirements, and the combat radius and sustained turn rate meet the desired values. The sizing plot (Fig. 2) for the air-to-air mission shows how the design point sits on the boundary of the feasible region, as indicated in the carpet plot (Fig. 1). The sizing plot design point has a takeoff wing loading of 96.86 lb/ft^2 and a maximum takeoff thrust-to-weight ratio of 0.95.

The results of the sizing trade, namely the wingspan and aspect ratio were subsequently used for the planform trade. Additional analysis is required to revalidate the new design, since the current design uses geometric values for the preliminary design to inform the weight and drag polar estimation of traded designs.

1.2 Planform Trade

After the design was resized using the refined methods, the planform trade from Assignment 6 was redone with additional fidelity. The updated AVL [1] file includes the wing, a model of the fuselage projected on the x-y plane, and the horizontal and vertical tails. The trade was done at the mid-mission cruise C_ℓ of 0.4323, with the objective to minimize cruise drag.

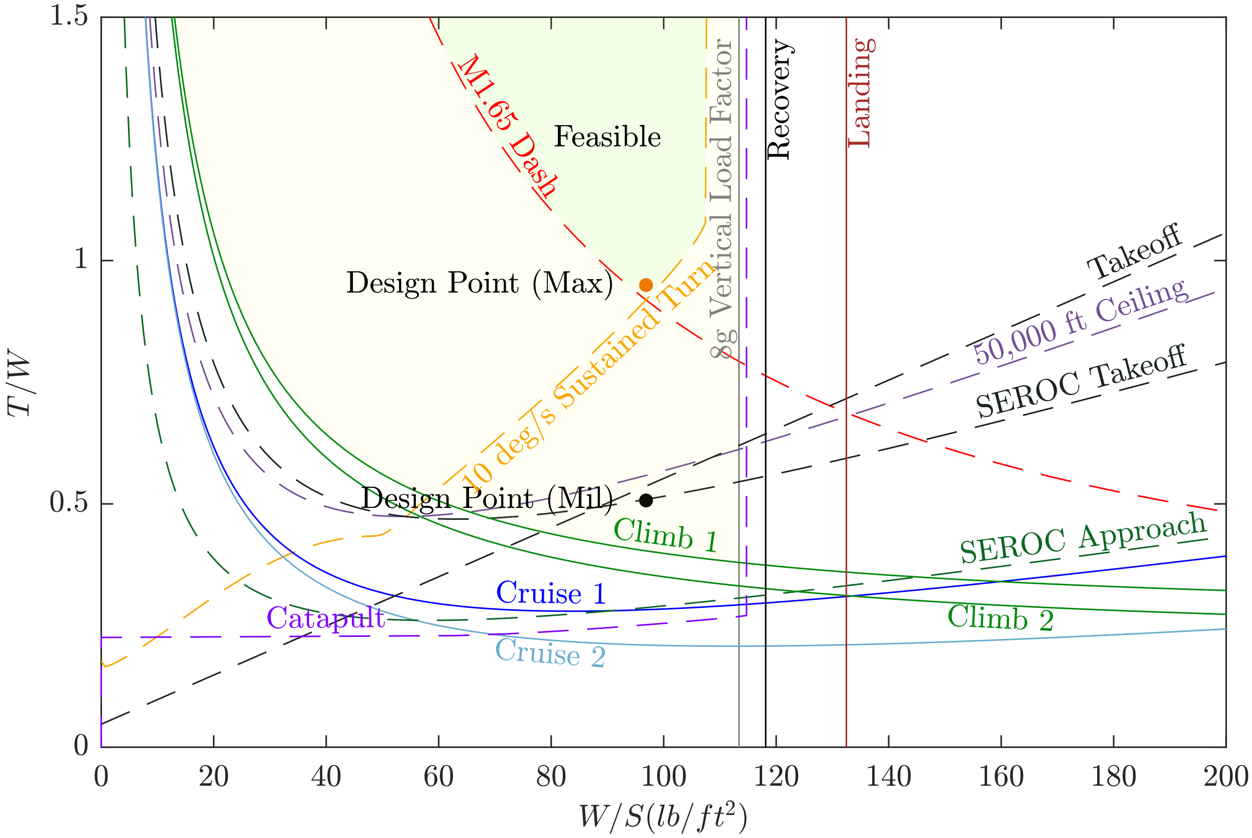

Figure 2: The resulting sizing plot from the sizing trade study.

The design variables changed were the wing taper ratio (λ), wing quarter-chord sweep angle ($\Lambda_{0.25}$), wing twist angle (θ_{twist}), wing x-location from the nose (x_{wing}), horizontal stabilizer moment arm length (ℓ_{HT}).

The C_D was modeled in a similar way to Assignment 6. The AVL output `CDff` gives the most accurate induced drag, which includes trim drag as the model is trimmed to $C_m = 0$. The `CDCL` keyword is used to model the parasite drag coefficient (C_{D_p}) of the wing. Equation 1, which is sourced from Raymer [2], is used to model the C_{D_p} of each of the tails. The S_{wet} is assumed to be twice the tail reference area, and S_{ref} is the wing reference area for correct normalization.

$$C_{D_p} = \frac{C_f \, FF \, Q S_{\text{wet}}}{S_{\text{ref}}} \tag{1}$$

Because of the large number of design variables, some assumptions were made to simplify the trade process. First, root incidence was set to zero, which is typical for fighter aircraft according to Raymer. Second, the CG location was assumed to be fixed in order to determine an x_{wing} that gives near neutral stability. Lastly, the optimal $\Lambda_{0.25}$ was assumed to be 30 deg, the lowest which results in zero wave drag, in order to increase $C_{L,\max}$.

An iterative process was used anto find the optimal θ_{twist} for each λ. There are two constraints, first that the static margin is near zero, and second that the wing and horizontal tail do not overlap. The horizontal tail was placed as close to the wing as possible to meet the 50 ft length constraint.

The design was iterated upon manually using an Excel tool that automatically interfaces with AVL using Visual Basic until the C_D was minimized. The resulting design has a λ of 0.2, a θ_{twist} of -2.7 deg, an x_{wing}

of 12.58 ft from the nose, and a ℓ_{HT} of 18.5 ft.

1.3 Ordnance and External Fuel Tanks Trades

The impact on drag during cruise due to external ordnance carriage and external fuel tanks is quantified in Table 1. Pylons and racks are included for each configuration. One pylon holds one external fuel tank, 2 MK-83 JDAMS, or 3 AIM-120C's. A rack is used for each set of MK-83 JDAMS and AIM-120C's. AIM-9X's are placed on the wingtips. The configuration is assumed to be symmetrical to maintain a y C.G. location of zero. Using methods from Raymer [2], the change in drag due to each object is calculated.

Table 1: Change in C_{D_0} in drag counts for external carriage of ordnance and fuel tanks. Strike mission loadout with external fuel tanks shows the highest additional drag.

	No External Fuel Tanks	2 External Fuel Tanks
No External Weapons	0.00	17.4
2 AIM-9X's (wingtip)	1.3	18.7
6 AIM-120C's	21.9	39.3
4 MK-83 JDAMS	22.9	40.3
Air-to-air	23.2	40.6
Strike	24.2	41.6

From the trade, it is clear that internal carriage of weapons and fuel is preferable. The least additional drag comes from the wingtip AIM-9X's, followed by the external fuel tanks. The strike mission loadout adds more additional drag than the air-to-air mission loadout. The strike mission ordnance with external fuel tanks increases the total drag of the aircraft at cruise by about 25%.

From Raymer [2], the additional drag from external weaponry and fuel tanks increases significantly in the transsonic and supersonic region. While this is not modeled here, the large increase in drag due to the externals will negatively affect mission performance. Since the aircraft is required to operate at higher Mach numbers for combat and dash mission segments, the internal carriage of weapons is highly preferred.

Compared to the potential external drag, the fuselage drag would have to be decreased up to 33% to counteract the additional drag of the externals. This is not feasible due to the engines taking up a fixed volume and the skin friction drag remaining relatively the same due to the overall length of the aircraft. The size and form factor of the fuselage would have to be drastically reduced to make such a significant reduction in drag possible.

Additionally, reducing external carriage is necessary for a reduced radar cross-section (RCS). External ordnance and fuel tanks are large contributors to the RCS of an aircraft [3]. Reducing RCS is possible by placing all weapons internally [4]. Additional factors that can reduce RCS include engine duct and inlet design, radar-absorbing materials (RAM), and fuselage shaping.

Placing ordnance and weapons internally is the most practical solution for drag reduction and aircraft survivability. However, the aircraft must still be capable of carrying 10,000 lbs externally. The F/A-67 will be capable of carrying external stores by utilizing removable pylons under the wings, similarly to the F-35 [5]. The aircraft will have the capability to carry additional stores for missions requiring more fuel or ordnance, but these will not be used for the air-to-air and strike mission to prioritize stealth and drag reduction.

2 Cost Analysis

The cost of the F/A-67 is found using a cost buildup from Roskam [6]. Roskam's manufacturing cost buildup already accounts for learning curve, where learning curves of 73-80% are typical for the airframe manufacturing industry.

2.1 Flyaway Cost

The flyaway cost is computed using the takeoff weight, tooling, materials, quality control, and labor costs from Roskam [6]. Labor and tooling rates are obtained from Raymer [2]. Raymer also provides the cost of avionics per pound. The flyaway cost is 98.2 $M 2025 per aircraft, which is 14.8 $M less than the F-18 unit cost of 113 $M 2025 [7]. Since the F/A-67 maximum takeoff weight is lower than the F/A-18E/F, the cost difference makes sense.

2.2 Direct Operating Cost

Direct operating cost is computed using fuel, oil, lubricants, personnel, weapons, and other operations and maintenance costs. The cost equations used come from Roskam [6]. The direct operating cost is 6.67 $M 2025 per year per aircraft.

2.3 Life-Cycle Cost

Life-cycle cost is the addition of Direct Operating Costs and Flyaway Costs to RDT&E cost. RDT&E cost is computed using equations from Roskam [6]. The life-cycle cost is 137 $B 2025 in total.

3 Group Work Split

Table 2: Work Split

Person	Work Percentage	Brief Description of Work Done
Michael Chen	16.67%	Did mission performance and sizing trade, updated performance parameters and sizing plot.
Charles Choi	16.67%	Redesigned fuselage and internals CAD, resized landing gear, gathered important parameters.
Cristina Erskine	16.66%	Did ordnance and fuel tanks trade, updated fuel weights, helped update cg and dimensions, helped write report.
James Gold	16.67%	Assisted with trade studies, conducted research on avionics, helped with CAD wetted areas.
Santiago Ramos-Assam	16.66%	Updated planform and control surfaces, obtained important measurements
Thomas Sheridan	16.67%	Updated lifting surfaces Excel sheet with tail and fuselage. Conducted aerodynamics trades, helped write report.

References

[1] *AVL 3.40*. 2022. URL: https://web.mit.edu/drela/Public/web/avl/avl_doc.txt.

[2] Daniel P. Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, 2018. ISBN: 9781624104909.

[3] Robert John Gilman. "Survivability Considerations During Aircraft Conceptual Design". MA thesis. Naval Postgraduate School, 1986.

[4] *Introduction to Radar Cross Section Reduction*. 2015. URL: https://assets.cambridge.org/97811070/92617/excerpt/9781107092617_excerpt.pdf.

[5] *External pylon training increases F-35A weapons payload capability*. 2017. URL: https://www.388fw.acc.af.mil/News/Article-Display/Article/1245390/external-pylon-training-increases-f-35a-weapons-payload-capability/.

[6] Jan Dr. Roskam. *Part 8: Airplane Cost Estimation: Design, Development, Manufacturing and Operation*. Roskam Aviation and Engineering Corporation, 1990.

[7] *F/A-18E/F Super Hornet Aircraft (F/A-18E/F)*. 2012. URL: https://www.globalsecurity.org/military/library/budget/fy2012/sar/f-a-18e-f_december_2012_sar.pdf (visited on 09/14/2025).